Adam McMillan

Melbourne, Victoria | +61 (0)403 680 684 | adam.j.mcmillan@gmail.com | linkedin.com/in/ajmcm

Software Engineering | Artificial Intelligence | Data | Research

Master of Information Technology (Artificial Intelligence) candidate at the University of Melbourne with an H1 average, Dean's Honours List recognition, commercial software development experience, machine-learning research experience and university teaching experience. Background spans software engineering, data analysis, AI/ML, research and client-facing development. Experienced in building and maintaining web applications, working with production data, developing and deploying machine-learning systems, conducting rigorous experimental research, and communicating technical ideas clearly.

Technical Skills

- **Programming:** Python, JavaScript, C++, Java, R, SQL
- **Software Development:** React, Django, HTML, CSS; front-end/back-end development, testing, deployment, maintenance and version control
- **Machine Learning & AI:** Scikit-learn, PyTorch, TensorFlow, Keras; classification, regression, neural networks, NLP, generative AI, model evaluation and fairness learning
- **Data & Analysis:** pandas, NumPy, SQL, data cleaning, relational databases, statistical analysis and reproducible experimentation
- **Cloud & Infrastructure:** Docker, Kubernetes, Google Cloud, Elastic, Fission, MPI, HPC
- **Generative AI Tools:** RAG, small-language-model fine-tuning, OpenAI and Anthropic APIs, Claude Code, Codex and LLM-based evaluation workflows
- **Collaboration:** GitHub, GitLab, technical documentation, university teaching and client communication

Professional Experience

Research Assistant (Software Engineering) – Griffith University, Queensland Conservatorium *Jul 2026 – Present*
Modernise a legacy university web application, investigate and resolve implementation issues, improve functionality and support delivery of a maintainable production-ready system.

Tutor – COMP90051 Statistical Machine Learning, University of Melbourne *2026 – Present*
Teach postgraduate students statistical machine learning through tutorials and practical exercises, supporting conceptual understanding, mathematical reasoning, Python/PyTorch implementation, debugging, model evaluation, assessment and feedback.

Software Developer (Casual) – Exergenics *Dec 2025 – Mar 2026*
Developed React and Django functionality in a technical building-energy optimisation environment. Built and deployed an internal machine-learning classifier, including data preparation, model development, evaluation and integration into an existing software workflow.

Software Developer Intern – Exergenics *Jul 2025 – Oct 2025*
Contributed to application features, maintenance, data cleaning, modelling and evaluation using operational HVAC data and Exergenics' in-house systems.

Freelance Developer – Self-employed *2020 – Present*
Develop and maintain web projects for small-business clients, handling requirements, implementation, testing, deployment and ongoing revisions.

Research

Latent-Group Methods for Exploratory Fair Classifier Learning *University of Melbourne, 2026*
Master's research project supervised by Jean Honorio; mark: 91 (H1). Developed and evaluated novel methods for identifying fairness-relevant latent audit groups in classification problems using Python and real-world datasets.

Exploring the Role of Motor Imagery in Disordered Movement in Children with ADHD *Deakin University, 2023*
Honours research in cognitive neuroscience involving experimental design, statistical analysis and interpretation of human-subject data.

Perfectionism in Classical Pianists: An Exploration of How It Manifests and Possible Coping Strategies *Griffith University, 2014*
Honours research examining perfectionism in classical pianists.

Education

Master of Information Technology (Artificial Intelligence) – University of Melbourne *2024 – 2026 (expected)*
Current graded-subject average: approximately 87.1 (H1); Dean's Honours List, 2025.
Selected results: Programming and Software Development 96; Research Project 91; Statistical Machine Learning 91; Database Systems & Information Modelling 91; Algorithms and Complexity 90; Natural Language Processing 85; Cluster and Cloud Computing 84; Software Processes and Management 84.

Bachelor of Psychology (Honours) – Deakin University, Cognitive Neuroscience Unit *2019 – 2023*
Class I Honours.

Bachelor of Music (Honours) – Griffith University, Queensland Conservatorium *2011 – 2014*
Class I Honours; University Medal.

Awards

- Dean's Honours List, 2025 – University of Melbourne
- Deakin Scholarship for Excellence – Deakin University
- Sir Samuel Griffith Scholarship – Griffith University
- University Medal – Griffith University
- 1st Prize, Great Romantics Competition – Melbourne Recital Centre
- 1st Prize, Queensland Piano Competition

References

Dr Jean Honorio
Senior Lecturer, University of Melbourne
Email: jean.honorio@unimelb.edu.au
Phone: Available on request
Relationship: Manager

Tony Sumongkol
Software Engineer, Exergenics
Phone: 0412 568 813
Relationship: Former colleague